

Beyond Accuracy

Mastering Classification Metrics (Precision, Recall, F1 & Multiclass)

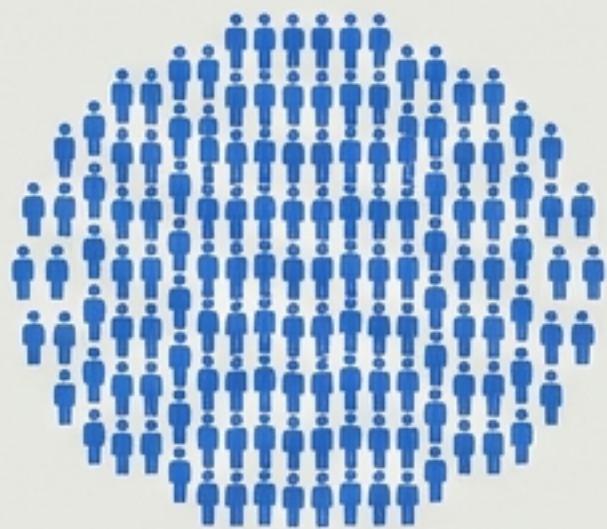


The Scenario



Scanning 1,000 passengers
to detect threats.

The Data



999 non-terrorists



1 terrorist

Actual

Predicted

		No Threat	Threat
Actual	No Threat	999 True Negative	0 False Positive
	Threat	1 False Negative	0 True Positive

$$999 / 1000 = 99.9\%$$

In imbalanced datasets, accuracy creates a dangerous illusion of success. The model is 99.9% accurate, but completely fails its objective.

Decoding the Errors

To move beyond accuracy, we must decide which type of error our business can afford to make.

Type 1 Error

False Positives – The False Alarm



Type 2 Error

False Negatives – The Fatal Miss



Precision and the Spam Filter Dilemma

The Scenario: Deploying a Gmail Spam Filter

The Risk



Out of everything the model predicted as positive, how many were truly positive?

Here, **False Positives** (flagging a genuine email as spam) are the **ultimate danger**. We need a model that guarantees a **positive prediction is actually positive**.

Calculating Precision

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Model A

True Positive ✓

100

False Positive ⚠

30

False Negatives

0

True Negatives

0

$$\frac{100}{100 + 30} = 77\%$$

Model B

True Positive ✓

100

False Positive ⚠

10

False Negatives

0

True Negatives

0

$$\frac{100}{100 + 10} = 91\%$$

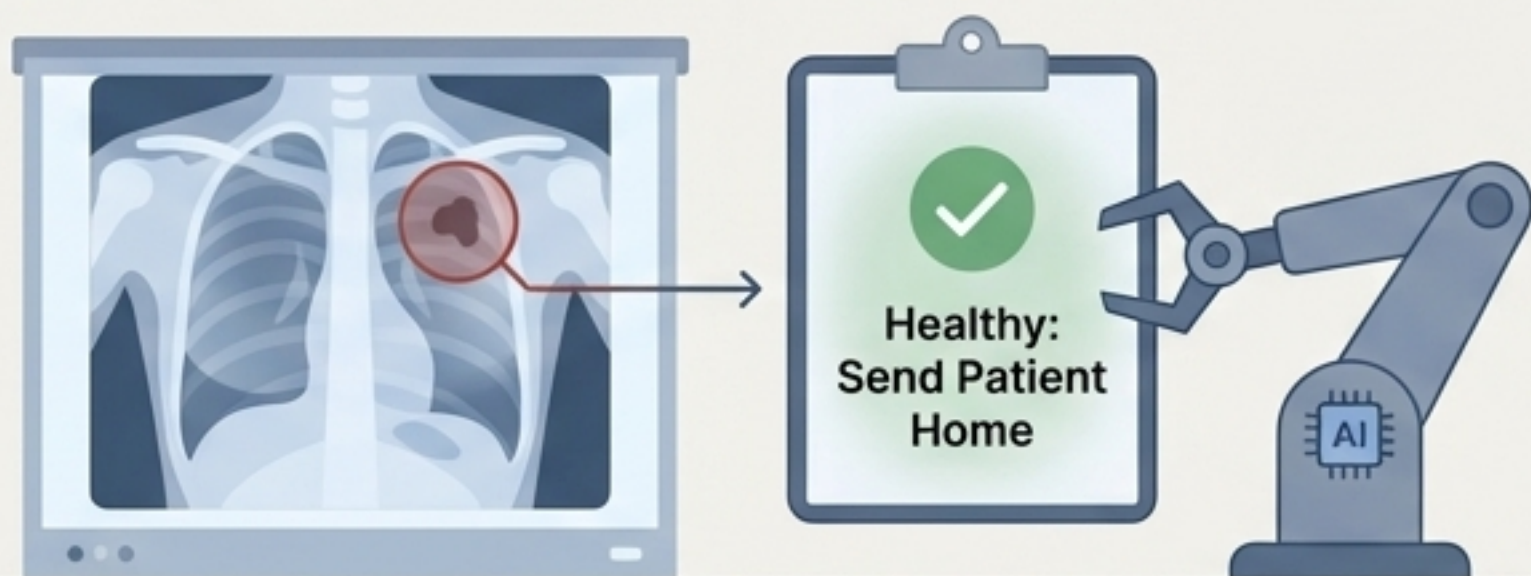


Model B wins. As a manager, you deploy the model that minimises Type 1 errors.

Recall and the Hidden Diagnosis

The Scenario:
AI for Chest X-Ray Diagnostics

The Risk



Out of all the **actual positive cases** in reality, how many did the model **successfully find**?

Here, **False Negatives** (missing a real diagnosis) are the ultimate danger. The patient will go about their life untreated. We need a model that **casts a wide net**.

Calculating Recall

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives} \text{ !}}$$

Model A

True Positive ✓ 1000	False Positive ⚠ 0
False Negatives ⚠ 200	True Negative ● 0

$$\frac{1000}{1000 + 200} = 83\% \text{ 🏆}$$

Model B

True Positive ✓ 1000	False Positive ⚠ 0
False Negatives ⚠ 500	True Negative ● 0

$$\frac{1000}{1000 + 500} = 66\%$$

Model A wins. When missing a positive case is unacceptable, you optimise for Recall.

The Precision-Recall Tension

The Cat vs. Dog Classifier: Calling a cat a dog (Type 1) is just as bad as calling a dog a cat (Type 2).



If you tweak a model to increase Precision (being overly cautious), Recall naturally drops. If you increase Recall (casting a wider net), Precision drops. You cannot maximise both simultaneously.

Finding Balance with the F1 Score

The Harmonic Mean

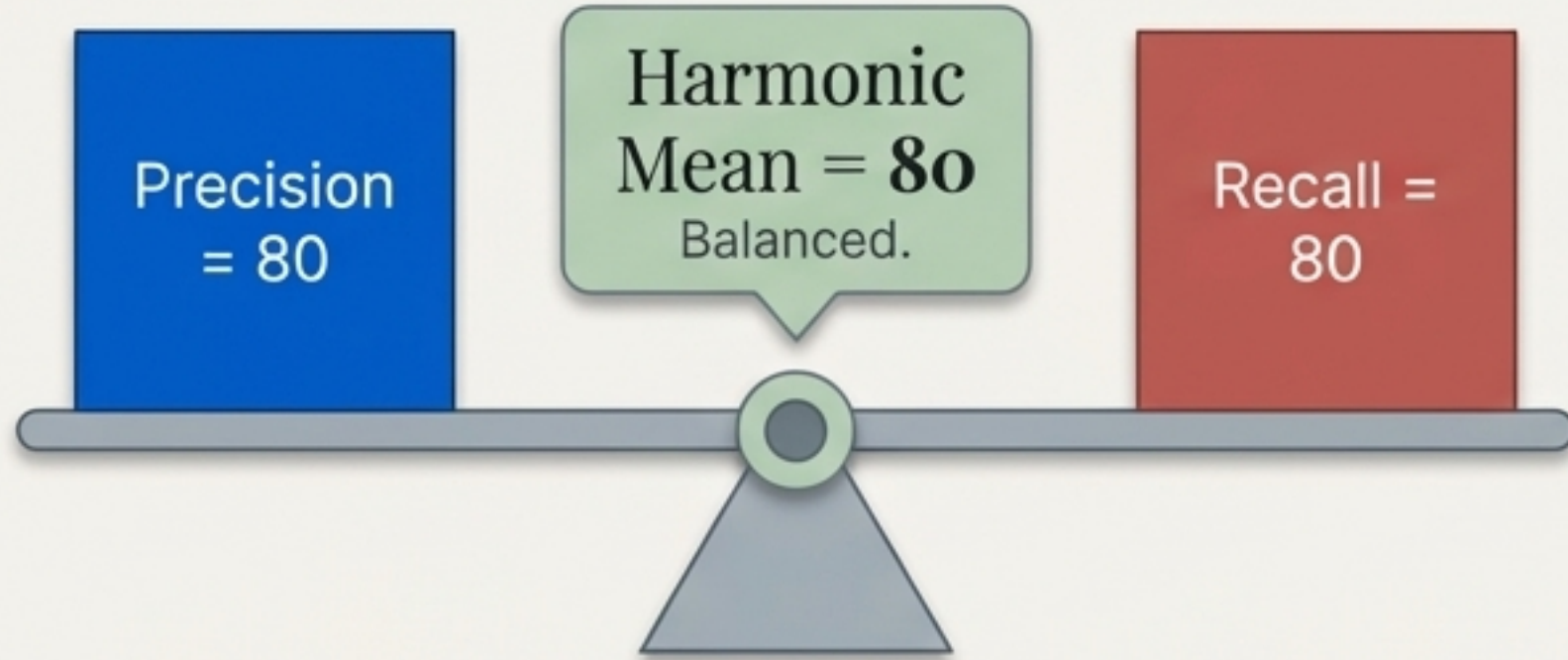
$$2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

When Type 1 and Type 2 errors carry equal weight, we need a unified metric. The F1 score merges Precision and Recall into a single number to evaluate overall model health.

The Harmonic Penalty

Scenario A

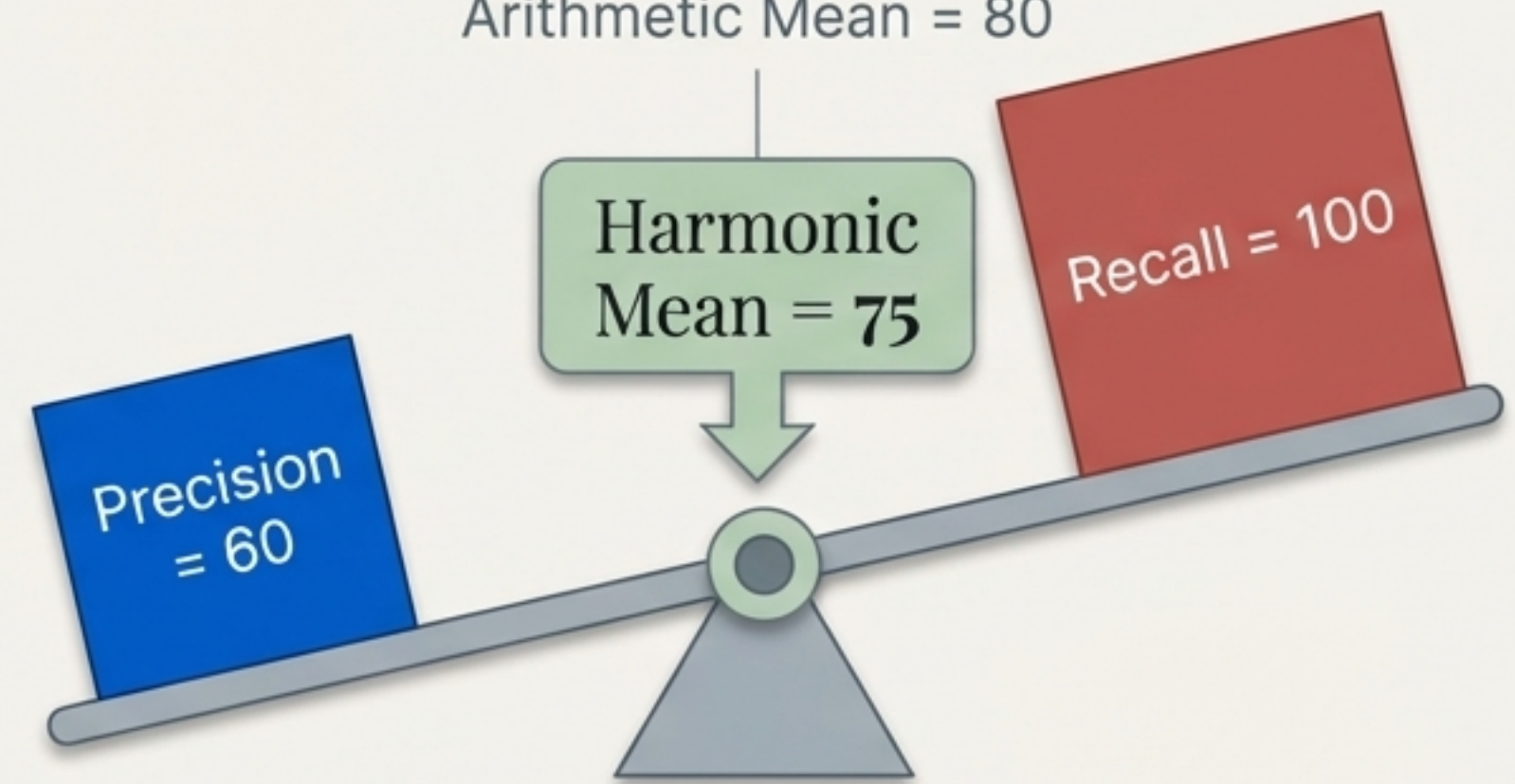
Inter



Scenario B

Inter

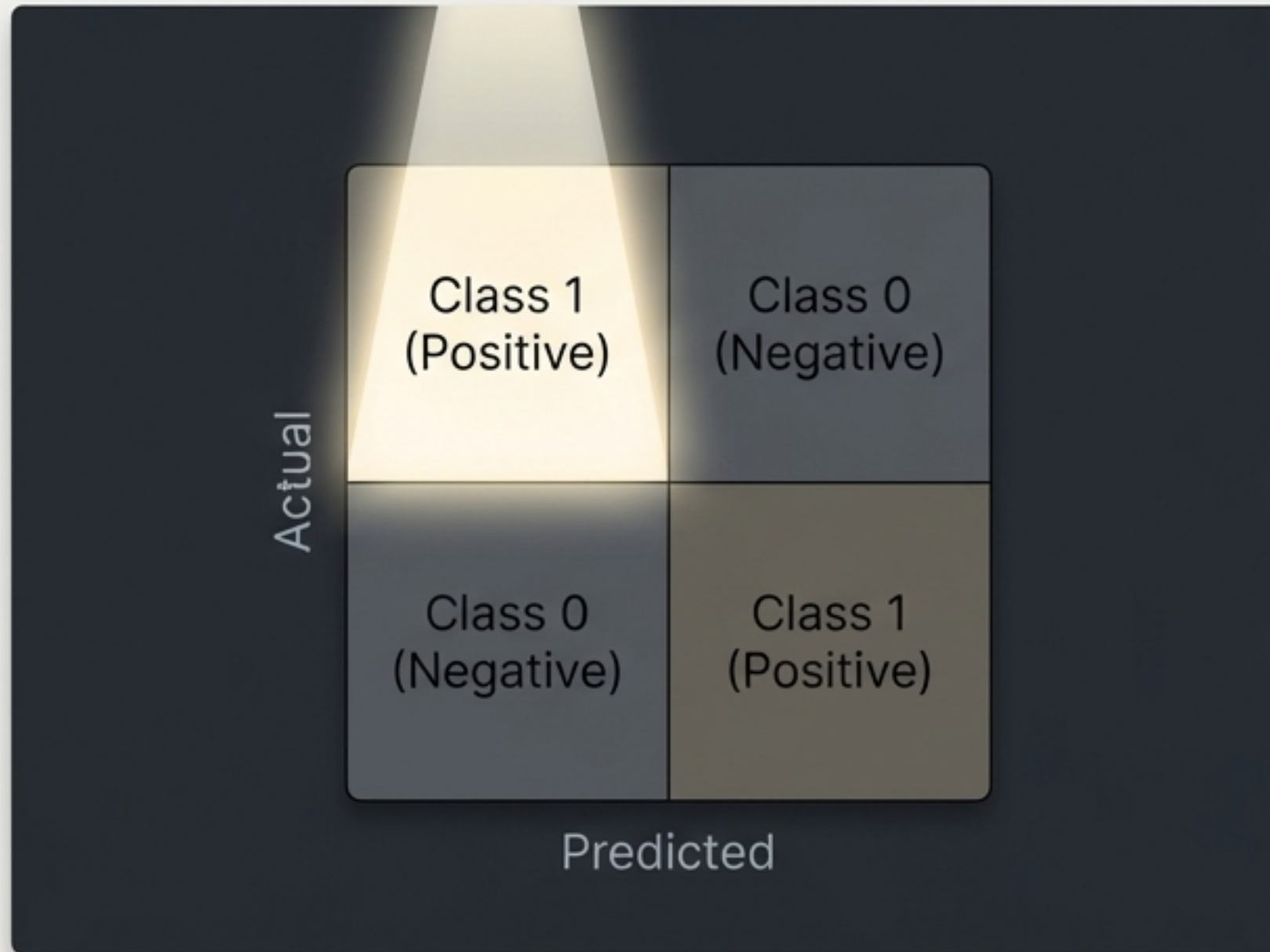
Arithmetic Mean = 80



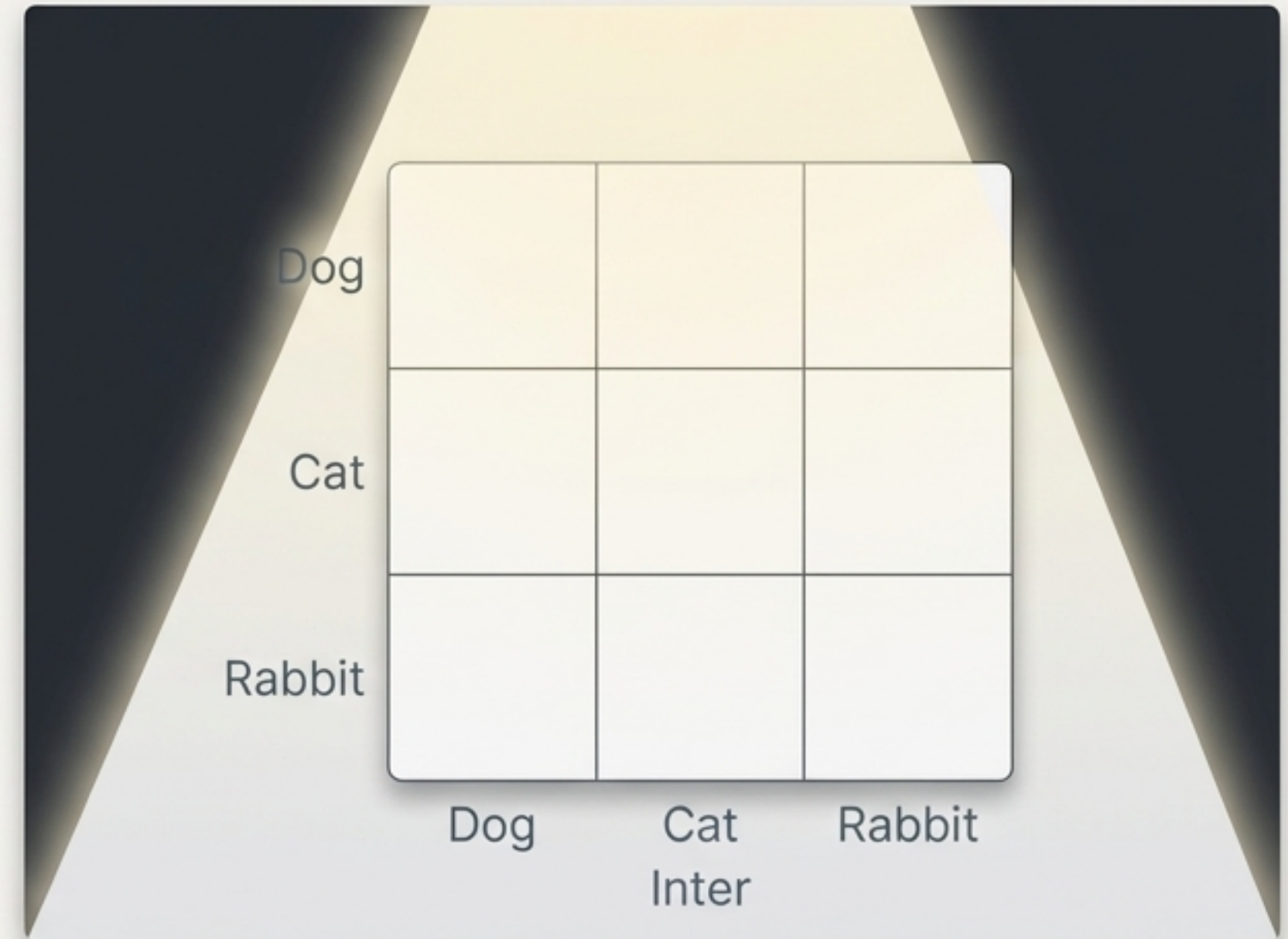
Unlike a standard average that sits perfectly in the middle, the Harmonic Mean penalises imbalance. It naturally pulls the final score closer to the lower value, forcing the model to be good at both.

Stepping Up to Multiclass Classification

Binary Classification



Multiclass Classification



Binary classification obsessively focuses on a single positive outcome. In multiclass environments, every label requires its own individual metric calculation before we can judge the model.

Deconstructing the Multiclass Matrix

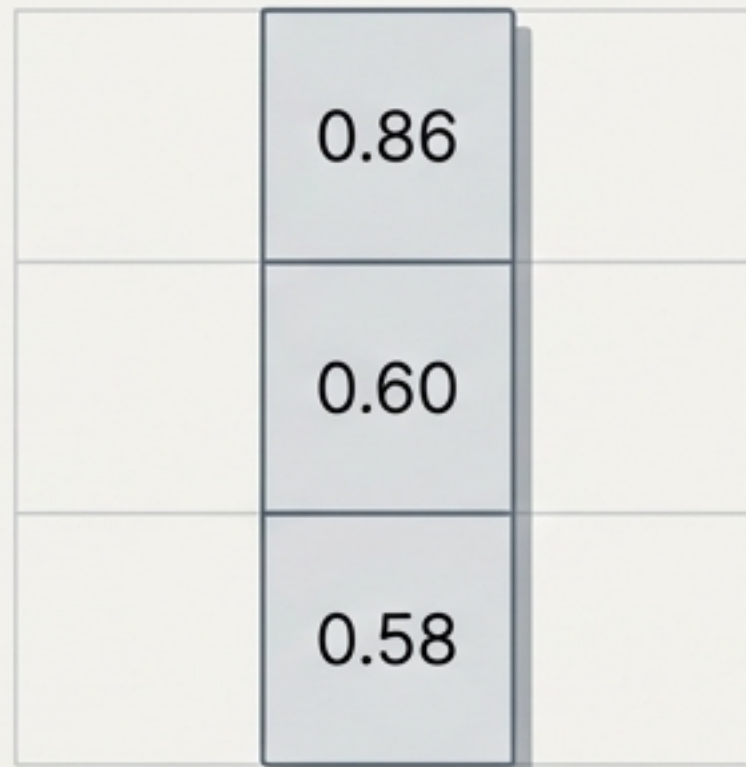
		Predicted Dog	Predicted Cat	Predicted Rabbit	Actual Totals	True Positives for each class
Actual	Actual Dog	29	0	0	Dog=29 Cat=45 Rabbit=34	
	Actual Cat	0	45	0		
	Actual Rabbit	0	0	34		
Predicted Totals		Dog=29	Cat=45	Rabbit=34		

To calculate Precision or Recall for 'Dog', isolate the green diagonal and divide by the respective orange column (Precision) or blue row (Recall).

Aggregating Multiclass Performance

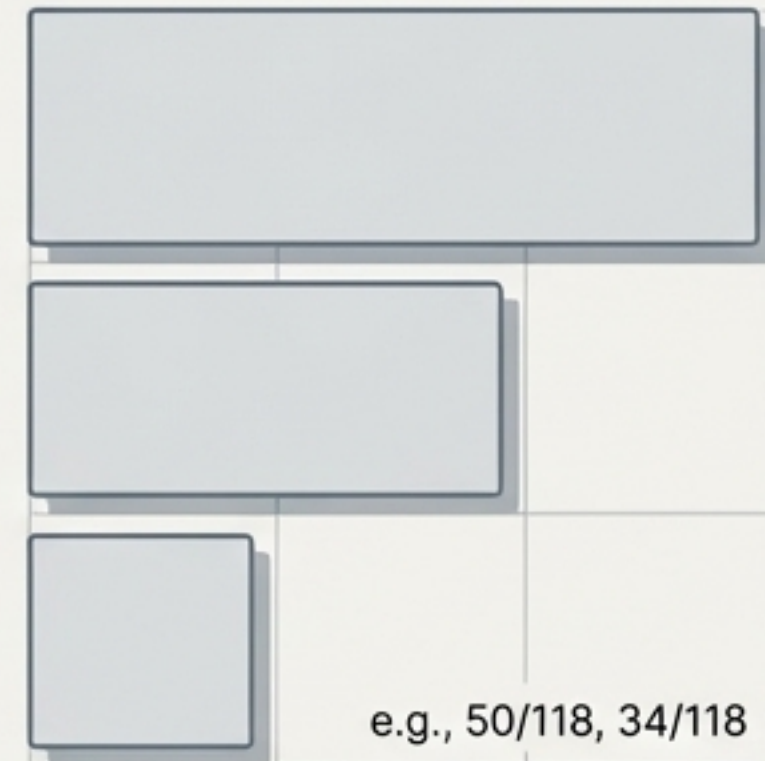
Class Scores: 0.86, 0.60, 0.58

Macro Average



$$\frac{(0.86 + 0.60 + 0.58)}{3}$$

Weighted Average



$$\text{Score} \times \frac{\text{Class Volume}}{\text{Total Data}}$$

Choosing the Right Aggregate

Is your dataset perfectly balanced?

YES

Use Macro Average

Treats all classes equally.
Best when no single class
dominates the data.

NO

Use Weighted Average

Accounts for class frequency.
Prevents a highly accurate majority
class from masking terrible
performance on a rare class.

The Practitioner's Toolkit

```
from sklearn.metrics import classification_report  
  
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.94	0.91	0.92	118
1	0.89	0.90	0.89	102
2	0.92	0.94	0.93	95
3	0.91	0.93	0.92	88
4	0.95	0.92	0.93	105
5	0.88	0.91	0.89	98
6	0.90	0.89	0.90	110
7	0.93	0.95	0.94	100
8	0.87	0.88	0.87	85
9	0.96	0.97	0.96	112
accuracy			0.92	1013
macro avg	0.91	0.92	0.92	1013
weighted avg	0.92	0.92	0.92	1013

A single line of code synthesises the entire theoretical journey. You now possess the context to read beyond the accuracy score.